

From Riemann Sums to the Fundamental Theorem

Riemann sums, accumulation, and conclusions from the Fundamental Theorem

- Problems 1–3 warm up one idea each: a Riemann sum, an accumulated total, a derivative of an accumulation function. After that they are shuffled — decide which idea a problem is about before you start.
- An estimate and an exact value are different answers. Read carefully which one is wanted.
- Give exact values (fractions, radicals) unless a decimal is requested.

1. Estimate $\int_0^2 x^2 dx$ with a *right* Riemann sum using 4 equal subintervals. Give the exact value of the sum.

Answer: _____

2. Water enters a tank at $r(t) = 4t + 3$ litres per minute. Find the total volume that enters during the first 6 minutes.

Answer: _____ L

3. Let $G(x) = \int_2^x (t^3 - 1) dt$. Find $G'(3)$.

Answer: _____

4. A cart's velocity is $v(t) = 3t^2$ metres per second. Find its displacement between $t = 1$ and $t = 4$ seconds.

Answer: _____ m

5. Consider $\int_1^3 \frac{1}{x} dx$.

- (a) Estimate it with a *left* Riemann sum using 4 equal subintervals. Give the exact value of the sum.

Answer: _____

- (b) Explain whether that sum is an overestimate or an underestimate.

Answer: _____

6. Let $H(x) = \int_1^{x^3} (2t + 1) dt$. Find $H'(2)$.

Answer: _____

7. A cart's speed is recorded every 2 seconds:

t (s)	0	2	4	6	8
$v(t)$ (m/s)	0	5	9	12	14

- (a) Use a trapezoidal sum over the four subintervals to estimate $\int_0^8 v(t) dt$.

Answer: _____

- (b) Say what units your estimate carries and what it measures.

Answer: _____

8. A reservoir changes at the net rate $r(t) = 10 - 2t$ thousand litres per hour for $0 \leq t \leq 8$, and holds 40 thousand litres at $t = 0$.

(a) Find the net change over the eight hours.

Answer: _____

(b) Find the amount held at $t = 8$.

Answer: _____

(c) Find the time at which the reservoir holds the most.

Answer: _____

9. Let $F(x) = \int_0^x f(t) dt$, where $f(t) = t^2 - 6t + 8$, on the interval $0 \leq x \leq 6$.

(a) Find the x -values at which F has a local extremum.

Answer: _____

(b) Justify which of them gives a local maximum of F .

Answer: _____

10. A tank fills at $r(t) = 6\sqrt{t}$ litres per minute for $0 \leq t \leq 9$, and $A(x) = \int_0^x 6\sqrt{t} \, dt$.

(a) Find $A(9)$.

Answer: _____ L

(b) Find $A'(4)$.

Answer: _____

(c) Explain how $A(9)$ and $A'(4)$ answer different questions about the tank.

Answer: _____